Reading Assignments: Multivariable Feedback Design

Multivariable Control, Homework 8

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This homework was a set of reading questions on course notes about multivariable feedback design. Each section ended with a summary of at most two pages. My summaries cover:

1. **Design objectives.** Good command following, disturbance rejection and tolerance to modelling error, written as conditions on the singular values of the sensitivity and complementary sensitivity functions. Also steady-state error and closed-loop stability.
2. **Normality.** How normal and non-normal plants affect performance, and modelling the plant through matrix decompositions.
3. **Model-based controllers.** Stability analysis in the time and frequency domains.

Each part gave practical conclusions on choosing loop shapes for MIMO controllers.